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Veripass- Passport forgery detection using CNN, OCR, and SIFT

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***Abstract*—The increasing occurrence of passport forgery presents a challenge to national security and the protocols used to verify identities. This research introduces a system for automated passport verification that combines Bilateral Convolutional Neural Networks (CNN) Optical Character Recog- nition (OCR) and Scale Invariant Feature Transform (SIFT). By examining the limitations of manual passport inspection methods and the growing complexity of forgery techniques this research sheds light on the need for advanced countermeasures. The proposed system revolutionizes the verification process, from document submission to approval resulting in improved efficiency and accuracy. With the use of Bilateral CNN for precise image preprocessing, OCR for text validation, and SIFT for image authentication this system provides a strong defense against passport forgery. This research addresses the requirement for automated forgery detection contributing to the advancement of passport verification systems. It emphasizes how crucial it is to maintain the integrity and security of identity verification processes in response to changing security threats. Index Terms—Passport forgery, Automated verification system, Bilateral Convolutional Neural Networks (CNN), Optical Char- acter Recognition (OCR), Scale Invariant Feature Transform (SIFT)**

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I. Introduction

The Forgery of passports presents a concern for the security of nations and the reliability of identity verification procedures. As forgery techniques become traditional systems for verifying Passports struggle to ensure the legitimacy of travel documents. This study aims to tackle these obstacles by suggesting a system for detecting passport forgery. It utilizes Bilateral Convolutional Neural Networks (CNN): Optical Character Recognition (OCR) and Scale Invariant Feature Transform (SIFT) to enhance the effectiveness of identifying passports.

A. Passport Forgery Challenges and Concerns:

Passport forgery refers to the replication or modification of passport documents enabling individuals to falsify their identities illegally cross borders or engage in activities. The conventional methods of verifying passports heavily rely on inspection, which is susceptible, to human errors and increasedingly sophisticated forgery techniques.

B. Limitations of Manual Passport Inspection:

The process of examining passport documents is time consumming and prone to errors particularly when dealing with a the volume of travelers.

C. Advancements, in Forgery Techniques:

Forgery experts continuously develop new methods to replycate authentic passports making it difficult for manual insection processes to keep up with these evolving forgery techniques.

D. Necessity of Passport Forgery Detection:

The detection of passport forgery is crucial, to address the dangers, it poses. It is imperative to develop an automated passport verification system that surpasses inspec- tion. This system should utilize state-of-the-art technologies to efficiently identify documents accurately.

E. Traditional System Workflow

The standard process of the system is as follows: The first step is the document summarization, in this process, individuals have to provide their details, along with the necessary documents, to apply for a passport. This ensures that all the important information is readily available for further processing. the second step is background checks, this is to ensure the safety and security of the passport issuance process, the authorities need to perform background checks. By doing this, they can identify any potential security concerns that

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might be linked to the applicant. To establish the authenticity of the individual, biometric data collection is carried out. This includes capturing their fingerprints and having them provide a passport-sized photograph. This de- tails play a crucial role in verifying their identity. When individuals cross the borders, it is the responsibility of the immigration officers to conduct manual inspections of their passports. The information and photographs provided by the individuals are thoroughly reviewed to ensure their identity. Once the verification process is completed, the next step involves stamping the passport. This signifies that the individual has been granted permission to either enter or exit the country. It's a significant milestone in the passport issuance process. While the traditional system has been, well, somewhat effective, the integration of the Bilateral CNN, OCR, and, uh, SIFT in our proposed passport forgery detection system aims to significantly, like, enhance the efficiency and accuracy of the verification process. The following sections will examine each component in-depth and show how their combination offers a complete answer to the problems caused by counterfeit passports.

II. Literature Survey:

In recent years, deep learning has gained prominence in image forgery detection. Syed sadaf Ali, Ngoch-son-vu introduced a novel approach that employs deep learning for image forgery detection by recompressing images. This technique can be adapted to identify manipulated passport images and stuff as passport forgeries often involves image manipulation[1]. P Zhou, X Han presented a method for learning rich features for image manipulation detection. While not passport-specific, this research is relevant for understanding the feature extraction and manipulation detection, which are vital components of passport forgery detection[2]. Our project uses Scale Inverse Feature Transform (SIFT) technology for feature matching. Jian Wul ,Ziming Cuil conducted a comparative study of SIFT and its variants. This research is fundamental to understanding the efficacy of SIFT, a key feature extraction technique in the context of passport forgery detection[3]. Chirag I. Patel and Atul Patel' paper explores the difficulties in text recovery from photos using optical character recognition (OCR), with a particular emphasis on Tesseract. It offers information on the history, uses, and particular difficulties faced by OCR. The study highlights the differences between Tesseract and Transym OCR and underlines how accurate Tesseract is at reading text from license plates. Overall, the study highlights Tesseract's special qualities and uses in OCR[4]. Pan and Lyu's work on detecting image region duplication, using SIFT features is pertinent to passport forgery detection. It addresses a specific manipulation technique often employed in passport forgery[5]. M Barni , M Nowroozi introduced a CNN-based approach for detecting generic contrast adjustment, with JPEG post-processing. This is directly applicable to detecting common manipulations in passport images, such as contrast adjustments[6]. L Bondi and S Tubaro discussed tampering detection and localization, which is relevant for localizing tampered regions in passport images, an essential aspect of forgery detection[7].M.J. Kwon ,H.K.Lee introduced CAT-Net, a network for the detection and localization of image splicing. This is directly relevant to passport forgery, where splicing is a common manipulation technique[8]. Jin's research on, like, image matching across wide baselines is relevant to matching genuine passport images, a crucial task in passport forgery detection[9]. Fakheri a framework for image retrieval using machine learning. This framework can be applied to retrieve reference passport images comparison during forgery detection[10]. Verdoliva's work on media forensics and deepfakes provides a broader perspective on image manipulation, which is valuable in understanding the broader context of passport forgery detection[11]. Yousfi and Fridrich's research on CNNs in JPEG steganalysis is essential in comprehending the challenges and potential solutions for detecting steganographic manipulations in passport images[12]. Mohan Goud Kath's research on Django, a totally robust web framework, provides stable development environment for the Scale-Invariant Feature Transform (SIFT) model in passport forgery detection. It offers a structured and scalable architecture, simplifies database management, and streamlines the integration of SIFT-based image processing [13]. Camacho and Wang's comprehensive review offers an overview of deep learning methods for image forensics, including their relevance to passport forgery detection.[14].Subramani, N., Matton' paper delves into the significance of automated document comprehension through deep learning, emphasizing challenges and the role of neural network models. It notes the scarcity of accessible datasets, relying on recent advancements in deep neural networks. The paper serves as a valuable resource for researchers exploring document comprehension methodologies[15]. Hamdi A, Linhares Pontes E, Sidere N paper explores how OCR technology affects the recognition and linking of named entities, in documents that have been OCRed. It offers an examination of the errors that can occur during OCR, such as character mistakes and problems with segmentation highlighting how these errors can impact the efficiency of indexing documents. The conclusion of the paper recommends implementing strategies for correcting OCR errors after the process emphasizing the significance of named entities, in libraries [16].

III. Proposed System:

In the given Fig 1, The architectural diagram illustrates the flow of operations within a comprehensive passport authentication system. At the forefront is the "User Interface," which provides a seamless avenue for users to submit their details via the "User Input" component. These inputs are then transmitted to the "Server-side Application," which is empowered by the robust Django Framework. Within the "Server-side Application," data is meticulously managed and stored in the "SQLite Database," ensuring very efficient and secure storage of user information. The heart of the system lies in the "Verification Services," which comprises multiple components such as "Image Processing," "Optical Character Recognition (OCR)," "Text Verification," and "Image Verifi-

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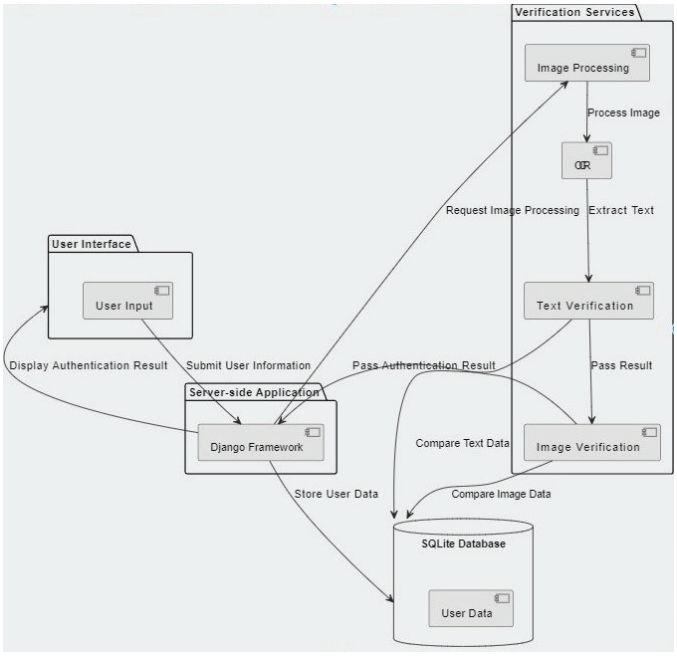


Fig. 1. Architecture of proposed system

cation." Image processing techniques enhance the quality of passport images, which are then subjected to OCR to extract textual information. Subsequently, text verification processes to compare the extracted text against the database are carried out for authenticity, while image verification techniques like, the Scale-Invariant Feature Transform (SIFT) ensure the validity of passport photos. Upon completion of these verification stages, the server-side application orchestrates the authentication process and communicates the results back to the user interface for seamless display to the end-user. This holistic approach ensures robust passport authentication, safeguarding against potential fraud or misuse.

In our comprehensive passport verification system, individuals willingly submit their details, including names, addresses, and photographs, to the designated authority. This is how the information is meticulously stored using the Django framework by leveraging an SQLite database for efficient and secure data management. When the need for verification arises, the authority initiates the process by inputting the passport front image into our system. This image then goes through a crucial preprocessing step where we use bilateral CNN. We enhance or optimize the image through smoothing and sharpening. The next process is Optical Character Recognition (OCR). It is a powerful tool. It meticulously does the job of extracting text characters from the passport image. We cross-reference the extracted text with the stored passport information text in the database. It's like playing a detective game, finding discrepancies and all. If things don't match up, indicating forgery potential, the system is all like "Hey! This passport is suspect! Investigate now!" Assuming successful text verification we move on to the image verification stage. We use the feature called the Scale-Invariant Feature Transform (SIFT). With SIFT, we compare the facial features in the passport photograph with the image stored in the database. If the image matches, image verification is done. But what if the images don't match? That means we've got ourselves a passport suspect. It adeptly identifies the passport as forged. Our system, for verifying passports emphasizes practices making sure to follow privacy regulations and handle data securely. We are committed to upholding the standards and treating individuals' information with respect and confidentiality. Our approach includes improvements enabling us to stay off emerging forgery techniques and incorporate user feedback to enhance our system continuously. In summary, our system is known for its strength, adaptability, and significant impact on improving security in passport verification processes setting benchmarks, in passport authentication.

IV. Methodology:

Image preprocessing is a step that involves applying techniques to raw images in order to improve their quality reduce noise and make them ready, for further analysis. For passport verification purposes;

A. Preparing the Image:

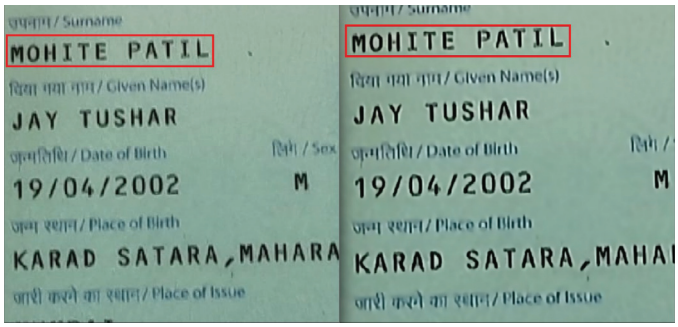


Fig. 2. Difference between a normal and a bilateraled image

In the above Fig 2, Utilizing Bilateral CNN; Convolutional Neural Networks (CNNs) are employed along with filtering to enhance the quality of the passport front image. Bilateral filtering helps in smoothing the image while preserving its edges. By utilizing CNNs we can significantly improve the clarity and representation of features in the image.

B. Combining OCR Text and SIFT Feature Extraction:

Optical Character Recognition (OCR): OCR is a technology that enables us to extract text from images. In this step, we utilize OCR algorithms to extract information like names and identification numbers from the passport image. OCR essentially provides us with a version of the text within the image.

Scale Invariant Feature Transform (SIFT) Feature Extraction: SIFT is a computer vision technique that allows us to extract features from images. It identifies points and descriptors that remain robust regardless of changes, in scale or

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orientation. In our case of passport verification SIFT helps us extract features from each passport image.

Fusion: The text extracted by OCR and the features obtained through SIFT are combined or fused. This combination results in a depiction of the visual details linked to each passport. The objective is to merge the advantages of OCR (text recognition) and SIFT (visual features) to enhance the verification process with reliability.

C. Mapping Text to Image Regions:

Text Regions Associated with OCR; The text extracted through OCR is connected to areas within the passport image. This process involves dividing the image into regions based on the identified text. Each region is then linked to information such as a person's name, identification number, or other significant details.

Establishing Systematic Mapping; A systematic mapping is created to establish an association between the OCR-mapped text regions and their corresponding textual information. This mapping ensures verification by connecting regions in the image with the extracted text.

D. SIFT Feature Matching:

During preprocessing SIFT features are extracted from the passport images. Form a reference feature set. These features act as a baseline for comparison during the verification process. In the Fig 3, a normal unfiltered image is matched with

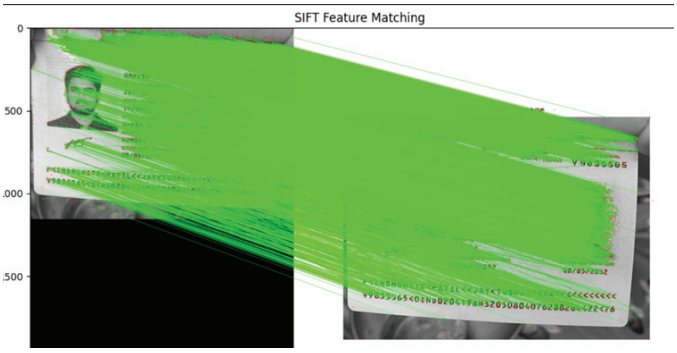


Fig. 3. SIFT feature matching using the unfilter image and the bilateral image yielding 7057 matched points

a bilateraled image to yield 7057 matched features.

When presented with a passport image for verification SIFT is utilized again to extract features. The matching of these features against the reference feature set involves assessing their similarity.

E. Verification and Authentication:

Utilizing Integrated Information; Verification utilizes both mapped text. Aligned SIFT features in combination. It combines details obtained through OCR with characteristics obtained through SIFT.

Setting Thresholds; Predefined thresholds, for matching accuracy are established. If both the written and visual elements align, within these limits the passport is considered genuine. However if there are inconsistencies that exceed the predetermined thresholds the passport is marked for scrutiny suggesting a case of forgery.

V. RESULTS AND DISCUSSIONS:

In the process of password verification, a multi-step approach is employed, combining Convolutional Neural Network (CNN) bilateral image processing, Optical Character Recognition (OCR) detection with string matching and Scale-Invariant Feature Transform (SIFT) feature detection. Each step contributed to the overall verification process.

A. CNN Bilateral Image Processing:

The initial step of the password verification process involved subjecting the input image to a sophisticated bilateral image processing using a trained Convolutional Neural Network (CNN). This advanced technique extracted and enhanced distinctive features present in the image, optimizing it for subsequent analysis. Through the application of deep learning algorithms, CNN effectively identified and highlighted relevant patterns, textures and structural elements within the image, thereby laying the foundation for accurate and reliable verification outcomes. The importance of bilateral filter- ing is that we can't only preserve the image quality but also improve it while reducing the image size. Fig 5 was originally 1 MB in size. To enhance it normally we would have increased its size and space complexity. CNN bilateral filtering greatly helped in achieving this.



Fig. 4. Original unedited image

B. OCR Detection and String Matching:

Following bilateral image processing, the processed image underwent meticulous Optical Character Recognition (OCR) detection. This intricate process involved the conversion of visual characters within the image into machine-readable text. Subsequently, the recognized text was meticulously matched against the expected password string for verification. By leveraging advanced OCR algorithms and sophisticated string-matching techniques, the system effectively scrutinized the extracted text ensuring precise alignment with the predefined password criteria. Similar technologies like OCR exist, name

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Fig. 5. The input image underwent bilateral image processing using a trained CNN. Distinctive features were extracted to enhance the image for subsequent analysis.

ICR and HTR, but our project emphasized accuracy and hence we chose OCR over the two. Among these, OCR stands out due to its versatility and accuracy. It is widely preferred for recognizing printed text from scanned documents, images , or printed material. OCR is the best when it comes to efficiently processing large volumes of printed documents. It can effortlessly handle various fonts, sizes, and styles. This makes OCR is an exceptional option for dealing with heterogeneous document types. Whether it's a formal document, a trendy magazine, or an ancient manuscript, OCR is up for the task. On the other hand, HTR specializes in understanding hand–written text. It is a niche technology with a narrower scope. While OCR triumphs over diverse text types, HTR focuses exclusively on handwritten content.

Similarly, ICR targets hand-printed characters in structured forms. Its application is restricted to processing structured documents with specific character formats. While this limitstation exists, ICR proves to be highly valuable in scenarios requiring accurate recognition of structured forms. OCR stands out for its high accuracy rates in extracting text from diverse document types and conditions, its versatility in adapting to different fonts and languages commonly encountered in the pass- ports, and its robustness in handling challenging scenarios such as low-quality images. In contrast, ICR is specifically designed for machine-printed text and may struggle with handwriting variations and noise in images, while HTR specializes in recognizing handwritten text and may face challenges with variations in handwriting styles and image quality. While OCR offers seamless integration with existing systems and work- flows. ICR and HTR may require additional preprocessing or specialized integration for their respective domains.

C. SIFT Feature Detection:

In the above Fig. 6, after matching the same given image with itself, the image matched a total of 5126 points or features, a total of 0 features remained unmatched, this shows that the image is completely authentic and identical. Our proposed

TABLE I
COMPARISON BETWEEN OCR, ICR, AND HTR

OCR	ICR	HTR
High accuracy, swift extraction for passport images.	lower accuracy with handwriting, processing speed varies on the quality of the image.	may struggle with machine-printed fonts, processing varies similarly.
Versatile and adaptable to different fonts, languages, and document formats and robust.	Limited adaptability to handwriting styles and struggles with noise.	Limited adaptability to machine-printed fonts and struggles with handwritting styles.

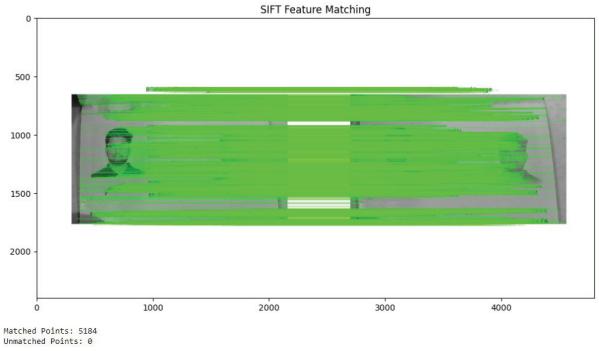


Fig. 6. Feature matching using SIFT technology

system uses SIFT technology for feature matching, another alternative available to us is SURF technology. In the field of computer vision and image processing, the SIFT (Scale-Invariant Feature Transform) and SURF (Speeded-Up Robust Features) algorithms play a significant role. Both algorithms aim to detect and describe features in images. However, they differ in their approach and computational efficiency. The SIFT algorithm detects key points in images using the identification of local maxima and minima of the difference of Gaussian (DoG) function in scale-space. The selection of key points depends on their stability across different scales and orientations. SIFT generates feature descriptors by utilizing gradient magnitude and orientation histograms in the local image patches that surround the key points. These descriptors are designed to be invariant to changes in scale, rotation, and illumination. Although SIFT is robust and effective in its functionality, it tends to be computationally more intensive due to its reliance on Gaussian convolution and gradient calculations at multiple scales. Similar to SIFT, the SURF algorithm detects key points using a similar approach. It employs the determinant of the Hessian matrix for scale and orientation selection. To achieve computational efficiency, SURF approximates the Gaussian second-order derivatives through the utilization of box filters. Feature descriptors in SURF are computed by utilizing Haar wavelet responses within fixed-size square regions centered on the key points. These descriptors are designed to be computationally efficient while maintaining robustness to scale and rotation changes. SURF unlike SIFT, is specifically designed for speed and

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efficiency. By approximating Gaussian derivatives with box filters and utilizing integral images for rapid convolution, SURF achieves faster computation as compared to SIFT. Particularly for our system, accuracy is the goal, so SIFT was preferred instead of SURF, though SURF has its advantages, a simple comparison between the two is given below.

TABLE II
COMPARISON BETWEEN SIFT AND SURF

Algorithm	Scale Rotation	Time	Memory needed	Matching
SIFT	Best Results	Slower	Larger	Best
SURF	Subpar Results	Faster	Smaller	Subpar

VI. CONCLUSION

This research tackles the problem of passport forgery by proposing a system that automates the verification process. By combining Bilateral Convolutional Neural Networks (CNN) Optical Character Recognition (OCR) and Scale Invariant Feature Transform (SIFT) the system provides a solution to improve the efficiency and accuracy of passport authentication. The research discusses the limitations of inspection methods and highlights the increasing sophistication of forgery techniques emphasizing the need for advanced measures of identity verification. The proposed system transforms the verification process resulting in enhanced security and reliability. Practical implications of this research include strengthening security measures improving border control procedures and reducing risks associated with passport forgery and fraudulent activities. In the proposed system we opted for SIFT over SURF due to SIFT's performance in extracting features especially when dealing with variations in scale and orientation. Similarly, we chose OCR, over Intelligent Character Recognition (ICR) and Handwritten Text Recognition (HTR) because OCR is versatile, accurate, and robust when it comes to extracting text from types of documents under conditions. These choices play a crucial role, in guaranteeing the effectiveness and the efficiency of our system for detecting forged passports. They greatly contribute to the improvement of passport verification systems and help safeguard identity verification processes against evolving security threats.

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